

Emily Stewart

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Education

Harvard University

Master of Science, Data Science (GPA: 3.93)

2024 – 2026

Cambridge, MA

Thesis: Developed methods to evaluate and improve LLM behavior in complex settings, focusing on learned evaluators and alignment with human judgments.

Washington University in St. Louis

Bachelor of Arts, Cognitive Neuroscience; Minor: Anthropology

2017 – 2021

St. Louis, MO

Phi Beta Kappa Honors Society

Experience

Harvard Research Fellowship at the Kempner Institute

Research Fellow | Advisors: Kianté Brantley, Mohit Bansal

Jun. 2025 – Aug. 2025

Cambridge, MA

Managed a series of integrated experiments to improve theory-of-mind abilities in LLMs for multi-turn dialogue.

- Designed and implemented a multi-agent reinforcement learning environment
- Developed modular training pipelines (SFT, PPO/GRPO, DPO) to systematically compare learning strategies for reasoning and agentic tasks
- Ran distributed training and inference on a SLURM-based GPU cluster, debugging training instability and maintaining reproducible workflows using containers and cutting edge RL frameworks for LLMs (SkyRL, Verifiers)

Computational Cognitive Neuroscience Lab, Harvard

Research Assistant | Advisor: Sam Gershman

Sep. 2024 – May 2025

Cambridge, MA

Developed and trained deep neural networks to study and replicate human physical reasoning about kinematics in simulation.

- Automated large-scale generation of simulation environments, improving efficiency of data collection and experimental analysis
- Benchmarked neural network predictions against human judgments collected through controlled behavioral experiments to evaluate alignment and generalization

The Syndemics Lab, Boston Medical Center

Research Data Analyst

Jul. 2022 – May 2024

Boston, MA

Managed several simulation modeling projects for the Syndemics Lab.

- Coordinated work among researchers, clinicians, and external partners, including CDC collaborators, and kept projects on track
- Led end-to-end Bayesian simulation studies projecting health and economic outcomes of substance use disorder and hepatitis C treatment policies under uncertainty
- Built and maintained R and SQL data pipelines that integrated with C++ simulation code for data cleaning, analyzing, and evaluating cost-effectiveness, population outcomes, and racial/ethnic disparities
- Resolved data gaps and validated model assumptions
- Translated model outputs into peer-reviewed publications, contributing to policy-relevant findings

The Syndemics Lab, Boston Medical Center
Research Program Specialist

Mar. 2021 – Jun. 2022
Boston, MA

- Led website, content creation, and social media strategy for lab, coordinating team deliverables
- Organized and coordinated the Opioid Modeling Consortium conference with Weill Cornell colleagues, overseeing speaker management, logistics, stakeholder meetings, and all deadlines
- Tracked and managed the writing and submission of grants and manuscripts

Johns Hopkins Medicine
Public Health Research Extern

May 2020 – Aug. 2020
Baltimore, MD

- Contributed to reports and a publication in the *Journal of Medical Ethics* on antibiotic resistance and hospital billing practices.
- Conducted literature reviews and web scraping for opioid prescribing policy projects.

Publications

Chatterjee A, Stewart EA, Assoumou SA, et al. Health and Economic Outcomes of Offering Buprenorphine in Homeless Shelters in Massachusetts. *JAMA Netw Open*. 2024 Oct 1;7(10):e2437233. doi: 10.1001/jamanetworkopen.2024.37233.

Hashim F, Hennayake S, Walsh CM, et al. Characteristics of US hospitals using extraordinary collections actions against patients for unpaid medical bills: a cross-sectional study. *BMJ Open*. 2022 Jul 12;12(7):e060501. doi: 10.1136/bmjopen-2021-060501.

Skills

Programming: Python, R, SQL

Machine Learning & LLMs: PyTorch, Hugging Face Transformers, scikit-learn, LLM APIs, LLM fine-tuning (SFT, DPO), RL for LLMs (SkyRL, Verifiers, GRPO / PPO)

Statistics & Research: experimental design, statistical modeling, Bayesian simulation, uncertainty quantification

Tools: Git, W&B, SLURM, reproducible workflows